Random Matrix Theory and Applications to Markowitz Portfolio Theory

A Report for
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Contents

1 Introduction and Preliminaries **2**
1.1 A short history of Random Matrix Theory 2
1.2 The Marchenko-Pastur Law 2

2 Current Literature on the Covariance Matrix **4**
2.1 Motivation: Portfolio optimization in Markowitz's theory 4
2.2 Covariance Cleaning Schemes 4
2.3 The Multi-Spiked Model and the BBP Phase Transition 7

3 An Alternative to the Sample Covariance Matrix **14**
3.1 Motivation: Fat Tails and the Failure of the standard Covariance Matrix 14
3.2 The Spartial Sign Covariance Matrix (SSCM) 15
3.3 Cleaning the Spatial Sign Covariance Matrix 17
3.4 A Convex Blend with the Gaussian-Cleaned Estimator 19

Abstract

In this paper, we investigate current random matrix theoretic techniques, specifically for applications in Markowitz portfolio theory. In Chapter 1, we introduce the Marchenko-Pastur Law, which is the limiting spectral distribution of a covariance matrix. Next, in Chapter 2, we look at current literature on applications to Markowitz Portfolio Theory. Specifically, we look at cleaning of covariance matrices and the Multi-Spiked Covariance Model. We also look into current literature on optimal shrinkage of eigenvalues outside of the Marchenko-Pastur bulk. Lastly, in Chapter 3, we develop an improved cleaning scheme that uses the Spatial Sign Covariance Matrix to build a portfolio that works well in turbulent markets.

1 Introduction and Preliminaries

1.1 A short history of Random Matrix Theory

Random Matrix Theory (RMT) is a field initially developed in 1928 by John Wishart when he introduced the notion of a random matrix. More specifically, he defined the probability distribution of the covariance matrix to be the *Wishart Distribution*. Now, in practice, the true covariance matrix is usually not known. Hence, this notion of a distribution for the covariance matrix is useful, as it allows us to use tools from multivariate statistics to be able to sample and estimate what the true covariance matrix is.

Then in 1950, physicist Eugene Wigner applied RMT to model the energy levels of heavy nuclei. In particular, he found a limiting distribution for the eigenvalues of certain large random matrices as the size increases. This result is not be important in this report. However, it demonstrates the notion of a limiting spectral distribution, which is incredibly useful as we are now able to then look at the eigenvalues and eigenvectors of a random matrix that we are interested in and study their asymptotic behavior. This brings us to the distribution that we are most interested in in this report — the *Marchenko-Pastur law*.

1.2 The Marchenko-Pastur Law

The Marchenko-Pastur law describes the asymptotic behavior of the eigenvalues of a particular matrix, similar to the Wigner Semicircle law mentioned previously. But instead of symmetric random matrices, the Marchenko-Pastur law describes the spectral behavior of *Wishart ensembles*.

Theorem 1.1 (Marchenko-Pastur Law)**.** *Let $\boldsymbol{X}$ be an $N \times T$ matrix of i.i.d. random variables with*

- $\mathbb{E}(X_{ij}) = 0$
- $Var(X_{ij}) = \sigma^2 < \infty$

Then, we construct the sample covariance matrix

$$\boldsymbol{C} = \frac{1}{T}\boldsymbol{X}\boldsymbol{X}^\top$$

We define $\lambda := \frac{N}{T}$ to be the dimension ratio of $\boldsymbol{X}$. As $N, T \to \infty$ such that λ converges to a deterministic limit $\lambda \in (0, \infty)$, the empirical spectral distribution (the distribution of the

eigenvalues) of $\boldsymbol{C}$ *converges almost surely to the Marchenko-Pastur distribution. The continuous portion of the probability density function for this limiting distribution is given by:*

$$f(x) = \frac{1}{2\pi\sigma^2\lambda x}\sqrt{(\lambda_+ - x)(x - \lambda_-)} \cdot \mathbb{I}_{[\lambda_-,\lambda_+]}(x)$$

where the upper and lower bounds of the eigenvalue bulk are defined as:

$$\lambda_\pm = \sigma^2(1 \pm \sqrt{\lambda})^2$$

Simulation. We generate a simple data matrix $\mathbf{X}$ with its entries $X_{ij} \sim \mathcal{N}(0, \sigma^2 = 1)$. We then compute a sample covariance matrix

$$\mathbf{S} = \frac{1}{T}\mathbf{X}\mathbf{X}^\top$$

Figure 1 shows the distribution of the eigenvalues of $\mathbf{S}$. Notice that all the eigenvalues are within the MP bulk, with the highest density being near 0. This plot aims to show the general shape of the Marchenko-Pastur Law given by Theorem 2.2.

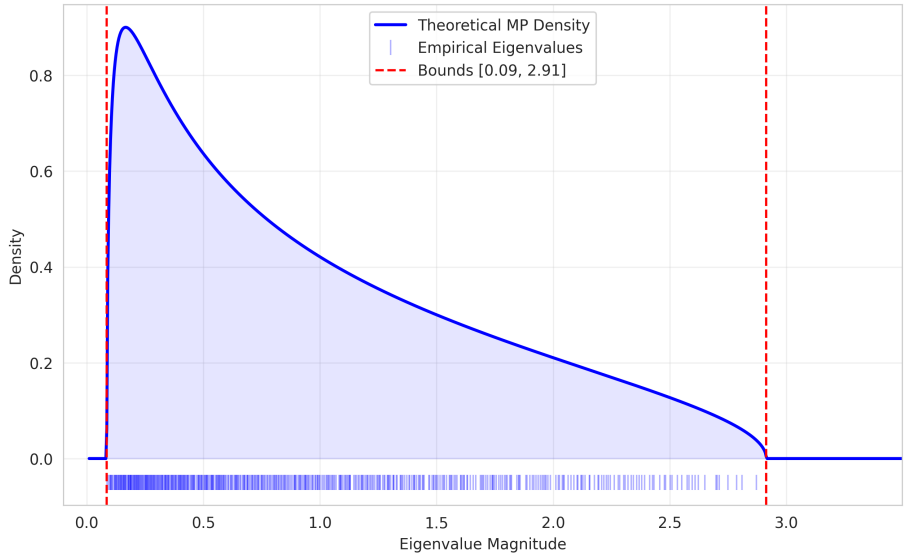


Figure 1: Visualization of the Marchenko-Pastur Law, $N, T = 500, 1000 \Rightarrow q = 0.5$,

2 Current Literature on the Covariance Matrix

2.1 Motivation: Portfolio optimization in Markowitz's theory

Say we have N assets, let Y_{it} be the return of the i-th asset at time $T = t$. Then we call $\mathbf{Y}$ the return matrix of size $N \times T$. One can then define the true covariance matrix to be $\mathbf{C}$, where

$$\langle Y_{it}, Y_{jt'} \rangle = C_{ij}\delta_{tt'}$$

that is, the correlation of 2 assets at different times is 0. Then, if we define $\mathbf{w}$ to be an N dimensional vector representing the weights of each of the N assets, one can minimize the variance to be:

Theorem 2.1 (Markowitz's portfolio optimization strategy)**.** *We claim that the solution to the following optimization problem*

$$\min_{\boldsymbol{w}\in\mathbb{R}^N} \frac{1}{2}\boldsymbol{w}^* \boldsymbol{C}\boldsymbol{w}$$

with constraint $w^T g \geq \mathcal{G}$ *is*

$$\boldsymbol{w_C} = \mathcal{G}\frac{\boldsymbol{C}^{-1}\boldsymbol{g}}{\boldsymbol{g}^T\boldsymbol{C}^{-1}\boldsymbol{g}}$$

where $\boldsymbol{g}$ *is the vector of predictions and* $\mathcal{G}$ *is the expected gains.*

Proof idea. The standard approach is to use Lagrange multipliers to solve this optimization problem with a constraint. □

This is what makes estimating the covariance matrix $\mathbf{C}$ important. Theoretically, if we are able to find a robust estimation for $\mathbf{C}$, we will be able to build a portfolio that is close to the theoretical ideal that minimizes volatility while maintaining the expected returns. And using the Marchenko-Pastur law in Chapter 1,we are able to investigate the empirical spectral density of sample covariance matrices, and by doing what is called *matrix cleaning*, we are able to find a sufficiently-good estimate for $\mathbf{C}$.

2.2 Covariance Cleaning Schemes

The motivation for cleaning covariance matrices is that we are then able to filter out real signal from noise. In linear algebra, eigenvalues are a measure of information. More specifically, when we look at the eigenvalues of a covariance matrix, say $\lambda_1 > \lambda_2 > \cdots > \lambda_n$, we are able to say something about the data. If $\lambda_1 \gg \lambda_2$, it means that the data is stretched out in the corresponding eigenvector's direction. This equates to a real structural *signal*. In quantitative finance, this represents the "market factor", which is the tendency of all the stocks to rise and fall together. However, when the eigenvalues are evenly spread out, that is, $\lambda_i \approx \lambda_j$ for $i \neq j$, then it means that the data has equal variance in all directions. This corresponds to noise.

It has been shown in Laloux et al. (1999) that the sample covariance matrices underestimates the smaller eigenvalues, and overestimates the larger ones. Hence, the raw sample matrix artificially distorts the boundary between signal and noise, exaggerating apparent structures while dangerously masking the random variance. Covariance cleaning schemes are therefore necessary to mathematically correct these biased eigenvalues, restoring the matrix so we can successfully isolate the true structural signal from the background noise. There are three notable cleaning schemes that many use in the context of portfolio optimization.

1. Linear Shrinkage
2. Eigenvalue Clipping

3. Rotationally Invariant Estimators (RIE)

We look into these three methods and evaluate their performance against real financial data.

1. Linear Shrinkage. Linear shrinkage is a regularization technique that constructs a well-conditioned estimator by pulling the empirical sample covariance matrix towards a highly structured target matrix. The fundamental intuition, formalized by Ledoit and Wolf (2004), is that the sample covariance matrix S is unbiased but highly sensitive to estimation error (high variance), whereas a structured target matrix F (such as the identity matrix or a constant-correlation matrix) contains significant bias but zero variance.

The linear shrinkage estimator Σ_{shrink} is defined as a convex combination of these two matrices:

$$\Sigma_{shrink} = (1 - \delta)S + \delta F$$

where $\delta \in [0, 1]$ is the shrinkage intensity. The optimal shrinkage intensity δ^* is derived analytically by minimizing the expected Frobenius norm between the shrinkage estimator and the true, unobservable covariance matrix Σ. By intelligently selecting δ^*, linear shrinkage guarantees a positive-definite, invertible matrix that systematically counteracts the eigenvalue dispersion problem, dragging the overestimated and underestimated sample eigenvalues toward a more stable central mean.

2. Eigenvalue Clipping. Eigenvalue clipping relies on the results of Random Matrix Theory (RMT) to establish a strict mathematical boundary between structural signal and random noise. Let X be a $T \times N$ matrix of random, independent, and identically distributed variables with variance σ^2. As $N, T \to \infty$ with a constant aspect ratio $q = N/T \in (0, 1]$, the Marčenko-Pastur theorem Marčenko and Pastur (1967) proves that the eigenvalues of the associated sample covariance matrix converge to a deterministic probability density function. Crucially, this distribution has a strict upper bound:

$$\lambda_+ = \sigma^2 \left(1 + \sqrt{q}\right)^2$$

In the context of financial data, any eigenvalue of the empirical correlation matrix that falls below λ_+ is theoretically indistinguishable from pure noise Laloux et al. (1999). Eigenvalue clipping addresses this by decomposing the sample matrix, retaining the "signal" eigenvalues ($\lambda_i > \lambda_+$), and replacing all "noise" eigenvalues ($\lambda_i \leq \lambda_+$) with a constant average value. This constant is specifically chosen such that the trace of the original covariance matrix is preserved, ensuring that the total variance of the system remains unchanged.

3. Rotationally Invariant Estimators (RIE). The Rotationally Invariant Estimator seeks to optimally clean the eigenvalues without altering the underlying eigenvectors. In high-dimensional regimes, the sample eigenvectors are known to overlap significantly with the true eigenvectors, meaning the "rotational" geometry of the sample matrix contains valid information even if the eigenvalues are heavily distorted.

Bun et al. (2017) demonstrated that the optimal RIE asymptotically minimizes the distance to the true covariance matrix under the constraint of rotational invariance. The cleaned eigenvalues λ_i^{RIE} are computed using the Stieltjes transform of the empirical eigenvalue distribution. Specifically, the RIE maps the empirical eigenvalues λ_i to their cleaned counterparts via a complex analytic function (the resolvent) of the sample covariance matrix. This provides a non-linear shrinkage effect that is dynamically dictated by the actual density of the sample eigenvalues, offering superior out-of-sample stability in portfolio optimization compared to standard linear shrinkage or hard clipping.

Analysis of cleaning methods. We proceed to look at the effectiveness of these cleaning methods in getting a robust estimation for the covariance matrix $\mathbf{C}$. We first get a sample

covariance matrix from an $N \times T$ data matrix of stocks from the $S\&P$ 500, with N being the number of stocks, and T represent the time (in days). Through data cleaning, N turns out to have a value of 491, and we choose to use data from `01/01/2023` $\sim$ `31/12/2025` (3 years), which gives $T = 750$. Therefore, our dimension of the data matrix $q \approx 0.65$.

Table 1: 2026 Out-of-Sample Performance Comparison

Cleaning method	**Ann. Return**	**Ann. Vol.**	**Sharpe Ratio**	**Max Drawdown**
S_1: Sample Covariance Matrix (SCM)	13.71%	13.06%	1.050	-10.48%
S_2: Linear Shrinkage	14.56%	10.43%	1.396	-7.79%
S_3: Eigenvalue Clipping	**24.37%**	9.55%	**2.552**	**-4.72%**
S_4: Shrinkage + Eigenvalue Hybrid	21.91%	**9.17%**	2.390	-4.87%
S_5: RIE	21.55%	9.68%	2.226	-5.57%
Equal Weight (1/N)	22.40%	12.28%	1.824	-7.60%

Results. It is important to note that the metric that is most important to compare effectiveness of cleaning is the *sharpe*, given by the following formula:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

where:

- R_p is the annualized expected return of the portfolio,
- R_f is the risk-free rate (often assumed to be 0 in out-of-sample relative performance comparisons),
- σ_p is the annualized standard deviation (volatility) of the portfolio's excess returns.

This metric is important because it rewards portfolios that achieve high returns with low volatility, which is what we desire in an *optimal* portfolio. In addition, one may also be interested in the Max Drawdown metic, which measures the difference between the highest and lowest return throughout the whole testing time period.

As seen in Table 1, Eigenvalue Clipping performed the best, achieving the highest sharpe ratio of 2.552 and the lowest Max. Drawdown. In general, we observed that S_3, S_4, and S_5 (main contenders in terms of matrix cleaning) clearly outperformed the rest, achieving a sharpe ratio of more than 2.0. This shows the effectiveness of these proposed methods in coming up with a closer portfolio to the ideal Global Minimum Variance Portfolio (GMVP).

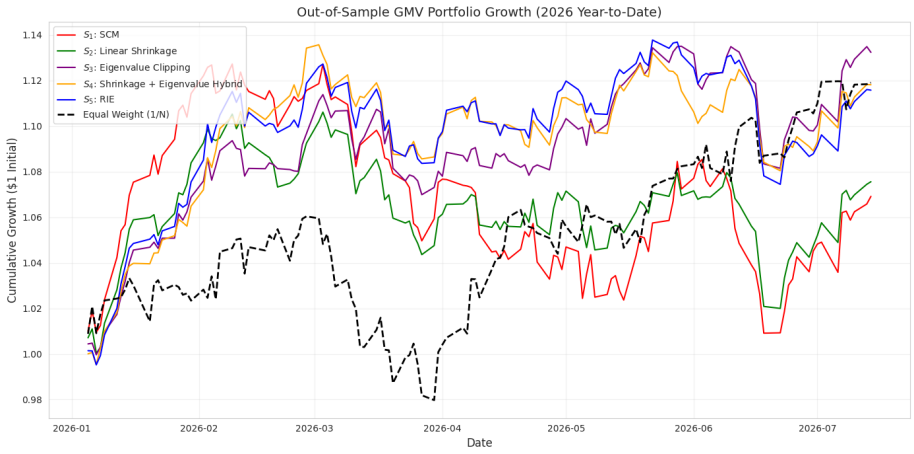


Figure 2: Plot of returns against the testing date (`2026-01` $\sim$ `2026-07`)

Although it may be irrelevant to the evaluation of the cleaning methods, Figure 2 provide a visualization of the effects of cleaning. Notice that for a portfolio of a higher variance (say S_1), the returns (y axis) is extremely volatile, that is, its movement are more "dramatic" compared to one where the covariance matrix was cleaned. (S_3 for example).

In addition, Figure 3 subtly shows us the effects of cleaning on the eigenvalues. S_1 denotes the sample covariance matrix. Hence, its spread of eigenvalues (in red) denotes the eigenvalues before cleaning. Notice that for all the cleaning methods (except clipping), the eigenvalues above the bulk can be seen to be shrunk. Linear Shrinkage (S_2) had a more aggressive shrinking compared to the other methods. And since eigenvalue clipping does not touch the eigenvalues above the MP bulk, the eigenvalues for S_3 are kept the same as S_1.

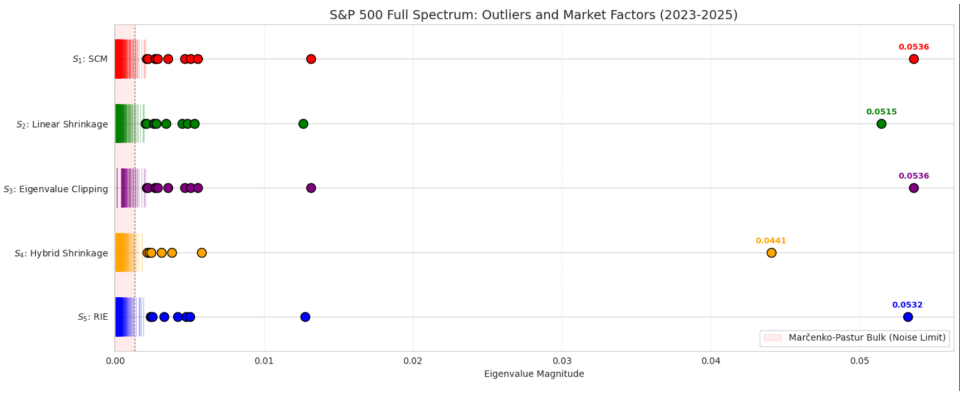


Figure 3: Spectral distribution of the 5 different matrices.

2.3 The Multi-Spiked Model and the BBP Phase Transition

The multi-spiked model was first developed by Johnstone (2001). He proposed that a covariance matrix comprised of two parts — signal and noise. In a standard pure noise model, $\mathbf{C}$ can be represented by the identity matrix (because we assume that there is no correlation between each stock). In a multi-spiked model, Johnstone proposed a *perturbation* of a finite number k (we call it a k-rank perturbation) of distinct eigenvalues (called the *spikes*) that are strictly greater than one. Mathematically, the eigenvalues of $\mathbf{C}$ that follows a multi-spiked covariance model could then be arranged as $\lambda_1 > \lambda_2 > \cdots > \lambda_k > 1 = \lambda_{k+1} = \cdots = \lambda_N$

And so, the covariance matrix under this model looked something like this:

$$\mathbf{\Sigma} = \mathbf{I} + \sum_{i=1}^{k} (\lambda_i - 1)\mathbf{u}_i\mathbf{u}_i^\top \quad \text{or equivalently,} \quad \mathbf{\Sigma} = \text{diag}(\lambda_1, \ldots, \lambda_k, 1 \ldots, 1)$$

In quantitative finance, these spikes are interpreted as market factors. And one would be interested in estimating these spikes. The pioneers of this problem was Baik et al. (2005). They found the asymptotic behavior of the spikes as $N, T \to \infty$ and $N/T \to q$ for some deterministic q. This result came to be known as the BBP Phase Transition. More formally,

Theorem 2.2 (BBP Phase Transition)**.** *Let $\boldsymbol{X}$ be an $N \times T$ data matrix. Suppose we have a multi-spiked covariance model (k-rank perturbation), where*

$$\mathbf{\Sigma} = \text{diag}(\lambda_1, \ldots, \lambda_k, 1 \ldots, 1)$$

with each $\lambda_i > 1$.

Then, let $\boldsymbol{S} = \frac{1}{T}\boldsymbol{X}\boldsymbol{X}^\top$. Let $\hat{\lambda}_1$ denote the largest eigenvalue of $\boldsymbol{S}$. The theorem asserts that as $N, T \to \infty$ such that the dimension of X, N/T converges to a deterministic limit, that is,

$$N, T \to \infty \quad \textit{such that} \quad N/T \to q \in (0, 1)$$

then the behaviour of $\hat{\lambda}_1$ *will go through the following phase transition depending on whether the true population spike* λ_1 *exceeds the threshold of* $1 + \sqrt{q}$*. Formally,*

$$\hat{\lambda}_1 \xrightarrow{a.s.} \begin{cases} (1+\sqrt{q})^2 & \text{if } \lambda_1 \leq 1 + \sqrt{q} \\ \lambda_1\left(1 + \dfrac{q}{\lambda_1 - 1}\right) & \text{if } \lambda_1 > 1 + \sqrt{q} \end{cases}$$

The BBP Phase Transition essentially says that the true structural signals (the spikes) can only be detected if their magnitude exceeds the critical threshold. If it is too weak (below the threshold), it gets absorbed into the Marchenko-Pastur bulk as noise. But if it is a sufficiently strong signal (above the threshold), then the sample eigenvalue will be separated from the MP bulk. In addition, the theorem gives us an exact expression for how much it gets separated. As observed by the expression, the sample eigenvalue $\hat{\lambda}_1 > \lambda_1$ at a high-dimensional regime. That is to say that the sample eigenvalues are overestimated. Hence, this gives us a glimpse of why cleaning regimes shown in Laloux et al. (1999) shrinks eigenvalues above the MP bulk.

Verification of the BBP Phase Transition. We run a simple simulation to verify the results claimed by Baik et al. (2005). We consider the following spiked covariance model (a 3-rank perturbation) of size N:

$$\mathbf{\Sigma} = \text{diag}(1.5, 1.8, 10, 1, \ldots, 1)$$

where there are $N - 3$ eigenvalues that are equal to 1.

The three spikes, $\{1.5, 1.8, 10\}$ are chosen intentionally to have two very close to the threshold of $1 + \sqrt{q}$ and one considerably far from it. This is so that we are able to capture the behavior of the eigenvalue for both cases. Next, we generate 500 $N \times T$ data matrices $\mathbf{X}_1, \ldots, \mathbf{X}_{500}$, and compute 500 sample covariance matrices:

$$\mathbf{S}_i = \frac{1}{T}\mathbf{X}_i\mathbf{X}_i^\top$$

We then consider a dimension of $q = 0.5$ (so that the threshold is $1 + \sqrt{q} \approx 1.707$), in order to observe all three cases:

1. When the spike is below the threshold (`spike` $= 1.5$)
2. When the spike is above (but close to) the threshold (`spike` $= 1.8$)
3. When the spike is above (and far from) the threshold (`spike` $= 10$)

As assumed in Theorem 2.2, the BBP phase transition holds only at a high asymptotic regime ($N, T \to \infty$). Therefore, it makes sense to increase N, T to be bigger, and observe any patterns in the distribution of eigenvalues. By keeping $q = 0.5$, we consider three regimes:

1. $N = 50, T = 100$
2. $N = 500, T = 1000$
3. $N = 1000, T = 2000$

Spike (l)	Asymptotic Limit	Value
10	$l\left(1+\frac{q}{l-1}\right)$	10.556
1.8	$l\left(1+\frac{q}{l-1}\right)$	2.925
1.5	$(1+\sqrt{q})^2$	2.914

Table 2: Asymptotic limits of sample top eigenvalues for a dimension ratio of $q = 0.5$.

Results. First, we calculate the theoretical asymptotic limits of the three spikes. This can be seen in Table 2. We then plot the 1500 eigenvalues (500 for each of the 3 spikes) in a plot as seen in Figure 4.

The red, blue and green dotted lines represent each spike's respective theoretical limit computed in Table 2. The histogram represents the distribution of the 1500 sample eigenvalues, with the red being the 500 largest eigenvalues, and the green being the second largest and finally the blue being the third largest. We can wee clearly that our results align well with Theorem 2.2. For $\hat{\lambda}_1^{(1)}, \ldots, \hat{\lambda}_1^{(500)}$, when they are put through the phase transition function $l\left(1+\frac{q}{l-1}\right)$, their values cluster around the theoretical BBP value of 10.556, denoted by the red line. A similar behaviour holds for the second largest spike, denoted in green. Another notable observation is that the variance of the distribution is clearly smaller for a higher-dimensional regime. That is, for larger values of N, T, the spread of eigenvalues is smaller compared to smaller values. This aligns with Theorem 2.2's claim that the asymptotic limit holds for very large N, T.

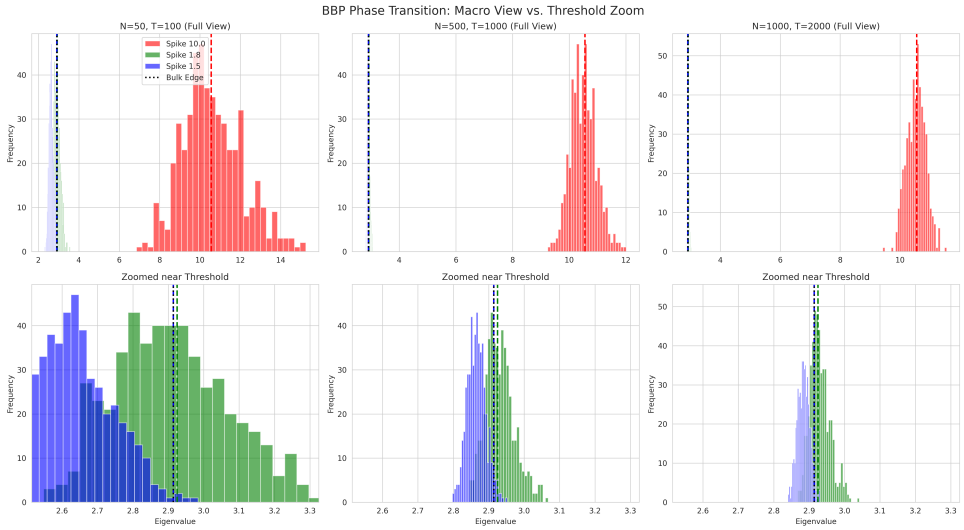


Figure 4: Distribution of Eigenvalues for a 3-spiked covariance model. $N, T = 50, 100, N, T = 500, 1000, N, T = 1000, 2000$ (from left to right)

What is interesting is the spike that was below the threshold doesn't cluster normally around the bound. However, an explanation for this would be due to a finite-dimensional regime. We notice that for $N, T = 1000, 2000$, its histogram is closer to the MP bound compared to that for $N, T = 500, 1000$ (and same for $N, T = 500, 1000$ compared to $N, .T = 50, 100$). We assume that as $N, T \to \infty$, the histogram collapses to a single peak at $\lambda = (1+\sqrt{0.5})^2$.

This simulation gives us confidence that the BBP Phase transition doesn't just hold theoretically, but it also holds in practice. It also gives us a way to predict the spectral behavior of a sample covariance matrix, which is extremely useful.

A suggested use of the BBP Phase Transition. A natural method to use the BBP Phase transition in constructing a shrinker function for the sample eigenvalues is

1. *Identify signal eigenvalues.* Take the sample eigenvalues $\hat{\lambda}_i$ such that $\hat{\lambda}_i > 1 + \sqrt{q}$ (the BBP threshold).

2. *Invert the BBP formulae.* For each detected signal eigenvalue ($\hat{\lambda}_i > 1 + \sqrt{q}$), estimate the true population spike l_i by taking the algebraic inverse of the BBP eigenvalue biasing function $\lambda(l) = l\left(1 + \frac{q}{l-1}\right)$. Solving this quadratic equation for l yields the debiased eigenvalue:

$$l(\hat{\lambda}_i) = \frac{(\hat{\lambda}_i + 1 - q) + \sqrt{(\hat{\lambda}_i + 1 - q)^2 - 4\hat{\lambda}_i}}{2} \tag{1}$$

3. *Collapse the noise.* For all remaining empirical eigenvalues $\lambda_i \leq 1 + \sqrt{q}$, apply eigenvalue clipping, that is, take the average of all eigenvalues in the MP bulk and take the average to be the noisy eigenvalues. In the multi-spiked covariance model, this average is simply 1.

4. *Reconstruct the covariance matrix.* Form the estimated covariance matrix $\hat{\mathbf{\Sigma}}$ by combining the debiased eigenvalues with their corresponding, unadjusted sample eigenvectors $\mathbf{u}_i$. Because we are ignoring eigenvector inconsistency, the original sample eigenbasis is strictly preserved:

$$\hat{\mathbf{\Sigma}} = \sum_{i:\lambda_i > 1+\sqrt{q}} l(\lambda_i)\mathbf{u}_i\mathbf{u}_i^\top + \sum_{i:\lambda_i \leq 1+\sqrt{q}} 1 \cdot \mathbf{u}_i\mathbf{u}_i^\top$$

However, a problem with this method is the inconsistency of the eigenvectors. Similar to the eigenvalues in the multi-spiked model, Paul (2007) found the asymptotic limit for the squared inner product of between the sample eigenvectors and its respective population eigenvector. Specficially, he found the asymptotic limit to be the following.

Theorem 2.3 (Paul (2007))**.** *Let $\boldsymbol{X}$ be an $N \times T$ data matrix, and let $\boldsymbol{S} = \frac{1}{T}\boldsymbol{X}\boldsymbol{X}^\top$ be the sample covariance matrix. Suppose we have the same multi-spiked covariance model (k-rank perturbation), with population spikes $\lambda_1, \ldots, \lambda_k$. Let $\mathbf{u}_i$ denote the unit population eigenvector corresponding to the true spike λ_i, and let $\hat{\mathbf{u}}_i$ denote the corresponding unit sample eigenvector of $\boldsymbol{S}$.*

As $N, T \to \infty$ such that the dimension ratio $N/T \to q \in (0, 1)$, the squared inner product converges almost surely as follows:

$$|\langle \hat{\mathbf{u}}_i, \mathbf{u}_i \rangle|^2 \xrightarrow{a.s.} \begin{cases} 0 & \text{if } \lambda_i \leq 1 + \sqrt{q} \\ \dfrac{1 - \dfrac{q}{(\lambda_i - 1)^2}}{1 + \dfrac{q}{\lambda_i - 1}} & \text{if } \lambda_i > 1 + \sqrt{q} \end{cases}$$

And one may notice that the asymptotic limit is less than 1. This is to say that the sample eigenvectors (in general) do not align well with the population ones.

Donoho et al. (2018) makes use of this fact. In his paper, he comes up with an optimal shrinker function that explicitly accounts for both the eigenvalue bias and this asymptotic eigenvector misalignment. To optimally correct for the discrepancies in the sample eigenvectors, the shrinker must penalize the eigenvalues *more* than what simple debiasing would dictate.

Specifically, the authors prove that there is no single universally optimal shrinker. The exact shape of the optimal shrinkage function, $\eta^*(\hat{\lambda}_i)$, is fundamentally determined by the *loss function* chosen to measure the discrepancy between the estimated and true covariance matrices. In fact, the authors show different optimal shrinkage functions for 26 different loss functions .

We choose to focus on the optimal shrinker for the *Stein loss* function for these two reasons.

1. *Precision Matrix Estimation.* Stein's loss is the canonical loss function for estimating the precision matrix ($\mathbf{\Sigma}^{-1}$), which is the fundamental object in constructing the GMVP.
2. *Error Maximization.* Sample covariance matrices, when inverted, exhibit "error maximization" where noise is interpreted as signal; Stein's loss specifically penalizes this behavior, making it highly robust for portfolio construction.

Under Stein's loss, the optimal shrinker is found to take a much more aggressive, non-linear form. Let $c(l(\hat{\lambda}_i)) = |\langle \hat{\mathbf{u}}_i, \mathbf{u}_i \rangle|$ denote the asymptotic cosine of the angle between the sample and population eigenvectors, and let $s(l(\hat{\lambda}_i)) = \sqrt{1 - c^2(l(\hat{\lambda}_i))}$ represent the corresponding asymptotic sine. The optimal shrinker is:

$$\eta^*_{St}(\hat{\lambda}_i) = \begin{cases} \dfrac{l(\hat{\lambda}_i)}{c^2(l(\hat{\lambda}_i)) + l(\hat{\lambda}_i)s^2(l(\hat{\lambda}_i))} & \text{if } \hat{\lambda}_i > 1 + \sqrt{q} \\ 1 & \text{if } \hat{\lambda}_i \leq 1 + \sqrt{q} \end{cases}$$

where $l(\hat{\lambda}_i)$ is the inverse BBP formula as shown in Equation 1.

And so, instead of trying to readjust the sample eigenvectors to their true directions, Donoho et al. (2018) instead solves an optimization problem to find an optimal shrinker that compensates for the persistent misalignment in the eigenvectors. Therefore, the fully optimized covariance estimator becomes:

$$\hat{\mathbf{\Sigma}}^* = \sum_{i:\hat{\lambda}_i > 1+\sqrt{q}} \eta^*(\hat{\lambda}_i)\mathbf{u}_i\mathbf{u}_i^\top + \sum_{i:\hat{\lambda}_i \leq 1+\sqrt{q}} 1 \cdot \mathbf{u}_i\mathbf{u}_i^\top \tag{2}$$

Analysis of the Stein-loss optimal shrinkage. We conduct 2 simulations to test the robustness of the Stein-loss optimised shrinkage function.

Simulation 1. In this first simulation, we run a simple Monte Carlo simulation to test whether the different construction gives a better loss

$$\text{loss} = \|\Sigma_{true} - \Sigma_{estimate}\|$$

The true covariance matrix Σ_{true} is chosen to be a multi-spiked model of three spikes, $\{5.0, 10.0, 15.0\}$. And the simulation parameters are chosen as follows:

1. $N = 100$, $T = 150, 300, 600, 1200$ (varying T)
2. $q \approx 0.33$
3. # of trials $= 2000$

For the 2000 trials, we generate 2000 different sample covariance matrices drawn from multivariate normal i.i.d. random variables. For each sample covariance matrix, we first clean the matrix via eigenvalue cleaning. Next, for the eigenvalues outside the bulk, we shrink them via the inverted BBP to get the first estimated matrix, and via the Stein-loss optimised shrinker to get the second estimated matrix. We then use the *Stein loss* to calculate 1000 losses for each of the 3 sampled covariance matrices $(\Sigma_{SCM}, \Sigma_{BBP}, \Sigma_{Stein})$ against the true covariance matrix that we constructed. We finally calculate the mean loss for each of the three matrices. We also computed the loss from using the *Frobenius loss*, just to see the results.

Results. Our results are summarised in Table 3 (Stein loss) and Table 4 (Frobenius loss). As expected, for the Stein loss (See Table 3), the Stein-optimal matrix always had a smaller loss compared to the BBP-inversed matrix. However, for the Frobenius loss (See Table 4, the BBP-inversed matrix had a smaller loss. This is to say that the Stein-optimal shrinker function is indeed optimised for the Stein loss.

Sample Size (T)	**SCM**	**BBP Inversion**	**Stein Optimal**
150	45.6112	1.8599	1.3733
300	19.0818	0.9709	0.8110
600	8.9179	0.4876	0.4423
1200	4.3269	0.2470	0.2331

Table 3: Stein Loss convergence as T increases.

Sample Size (T)	**SCM**	**BBP Inversion**	**Stein Optimal**
150	111.4220	45.2988	80.2843
300	55.2484	22.9702	39.7123
600	27.5144	11.5575	17.7626
1200	13.8132	5.8574	7.6326

Table 4: Frobenius Loss convergence as T increases.

Simulation 2. For the next simulation, we test the robustness of this method on financial data. We repeat the experiment that we did in Chapter 2.2 on the different cleaning methods. But instead of the four different cleaning methods, we use the (1) Sample Covariance Matrix, (2) Matrix obtained via inversion of the BBP (3) Matrix obtained via the construction given by Equation 2. We use the same training data (`2023-01` $\sim$ `2025-12`) and test it on the year 2026.

We get the sample covariance matrix via the data matrix pulled from financial data. Then, we use the cleaning pipeline we used in *Simlation 1* (eigenvalue clipping + respective shrinking factor for outside-of-bulk eigenvalues). Our $N, T = 491, 750$, giving us a dimension of $q = 0.65$.

Estimator ($q = 0.65$	**Ann. Return**	**Ann. Volatility**	**Sharpe Ratio**	**Max Drawdown**
SCM	10.31%	13.02%	0.792	-10.48%
BBP Inversion	24.40%	9.42%	2.591	-4.66%
Stein-Loss Optimal	25.16%	8.92%	2.819	-4.23%
Equal Weight (1/N)	22.70%	12.23%	1.856	-7.60%

Table 5: 2026 Out-of-Sample Performance Comparison for Global Minimum Variance Portfolios.

Results. As seen in Table 5, the Stein-loss optimal cleaning had the best Sharpe ratio. These results support the findings of Donoho et al. (2018), specifically that the BBP-inversion targets the correct eigenvalue but ignores the estimation error in the eigenvector itself, whereas Stein's loss penalizes exactly that misalignment, giving it an edge when the sample eigenvector overlap is not close to 1. Also, we can see Donoho et al. (2018)'s claims in Figure 5, where the shrinking intensity can be seen. For the Stein-optimal shrinkage, the shrinking is very intense, compared to a very subtle shrinkage in the inverted BBP. Another observation that we had, which we also saw for the linear and Rotationally Invariant Estimator (RIE) in Chapter 2.2, was that the larger eigenvalues had a larger shrinkage compared to the ones closer to the bulk.

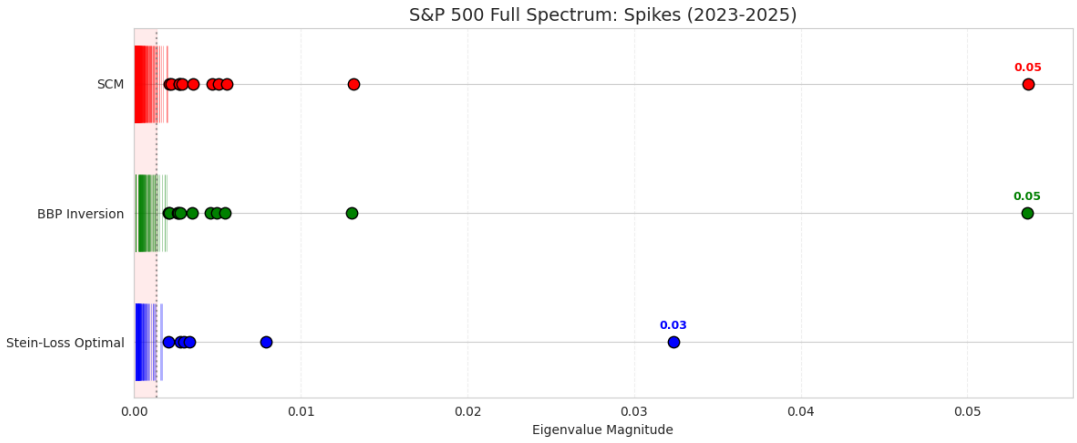


Figure 5: Distribution of Eigenvalues from SCM and the 2 different shrinking functions

3 An Alternative to the Sample Covariance Matrix

3.1 Motivation: Fat Tails and the Failure of the standard Covariance Matrix

The Marchenko-Pastur framework that we developed in Chapter 1 assumes that we have a finite fourth moment. However, in actual financial data, stock returns are known to exhibit *excess kurtosis*, that is, having fat-tails. (For a more comprehensive discussion on this topic, see Cont (2001))

To show the downside of using the standard covariance matrix in portfolio optimization, we consider a simple simulation. Consider an 3×10 data matrix (Number of stocks $= 3$, number of days $= 10$).

Example 3.1. Consider these two data matrices:

$$\mathbf{X}_1 = \begin{bmatrix} 1 & -1 & 2 & 0 & -1 & 1 & -2 & 1 & 0 & -1 \\ 0 & 2 & -1 & 1 & -2 & 0 & 1 & -1 & 2 & 0 \\ -1 & 0 & 1 & -2 & 1 & -1 & 0 & 2 & -1 & 1 \end{bmatrix}$$

$$\mathbf{X}_2 = \begin{bmatrix} 1 & -1 & 2 & 0 & -1 & 1 & \mathbf{-25} & 1 & 0 & -1 \\ 0 & 2 & -1 & 1 & -2 & 0 & \mathbf{-30} & -1 & 2 & 0 \\ -1 & 0 & 1 & -2 & 1 & -1 & \mathbf{-20} & 2 & -1 & 1 \end{bmatrix}$$

$\mathbf{X}_1$ models a *normal* market model, that is to say, a finite-variance regime. However, $\mathbf{X}_2$ models $\mathbf{X}_1$, just that on Day 7 (corresponding to Column 7 of $\mathbf{X}_2$), the stock returns considerably. And when we calculate the covariance matrix of both matrices via $\mathbf{\Sigma}_i = \frac{1}{10}\mathbf{X}_i\mathbf{X}_i^\top$, we get:

$$\mathbf{\Sigma_1} = \begin{bmatrix} 1.4 & -0.5 & 0 \\ -0.5 & 1.6 & -0.9 \\ 0 & -0.9 & 1.4 \end{bmatrix} \quad \mathbf{\Sigma_2} = \begin{bmatrix} 63.5 & 74.7 & 50.0 \\ 74.7 & 91.5 & 59.1 \\ 50.0 & 59.1 & 41.4 \end{bmatrix}$$

If we compute their weight, we find that:

$$\mathbf{w_1} = \begin{bmatrix} 0.253 \\ 0.384 \\ 0.363 \end{bmatrix} \qquad \mathbf{w_2} = \begin{bmatrix} -0.109 \\ -1.152 \\ 2.261 \end{bmatrix}$$

For $\mathbf{w}_1$, the GMVP weights are well-distributed. They recognise the signals from our data matrix, and allocate 25% to Asset 1, 38% to Asset 2, and 36% to Asset 3. However, $\mathbf{w}_2$ takes an extreme stance. Due to the simulated market crash in Day 7, the optimer tells us to take a long position for a whopping 200% on Asset 3, and short Asset 1 and 2 by 10.9% and 115.2%.

This demonstrates the downside of using a standard covariance matrix, that is, random market factors (such as market crashes during COVID or the 2008 Financial crisis) can cause the optimiser to think that there are real market signals between stocks, when in actuality, these are mere noise in the market.

This brings us to the motivation for this chapter. As mentioned, stock returns are highly affected not just by real market factors, but it can also be affected by noise. Noise can take the form of a market crash, or a sudden, isolated liquidity shock. Because standard covariance calculations rely on squared deviations, they are disproportionately sensitive to these heavy-tailed outliers. A single severe market event can permanently distort the empirical eigenvalue distribution, smearing the boundary between true signal and random noise. To resolve this, we suggest an estimator that evaluates the direction of asset returns independent of their magnitude, motivating the use of the Spatial Sign Covariance Matrix (SSCM).

3.2 The Spartial Sign Covariance Matrix (SSCM)

To construct the Spatial Sign Covariance Matrix, we must first define what we call the *spatial sign function.*

Definition 3.2 (Spatial sign function)**.** For a given N-dimensional return vector x_t at time t (that is, the t-th column of a data matrix), its spatial sign s_t is obtained by projecting the vector onto the unit hypersphere. Mathematically, this transformation is defined as:

$$s_t = S(x_t) = \begin{cases} \frac{x_t}{\|x_t\|} & \text{if } \|x_t\| \neq 0 \\ \mathbf{0} & \text{if } \|x_t\| = 0 \end{cases} \tag{3}$$

where $\|x_t\|$ denotes the Euclidean L_2–norm of the vector.

By applying this function, the magnitude of every non-zero observation is fixed exactly to 1. This geometric projection strips away the severity of extreme tail events, forcing the data into a strictly bounded support where all statistical moments, including the fourth moment required for random matrix edge limits, are guaranteed to be finite.

Then, we first transform the data matrix to a *spatial-sign data matrix*, which is basically a data matrix with its columns being transformed by the spatial sign function. More formally, given a data matrix $\mathbf{X}$, let $\mathbf{x}_t$ denote the t-th column of $\mathbf{X}$. Then, we form a new data matrix $\mathbf{S}$ with columns being $\mathbf{s}_i := S(\mathbf{x}_t)$. Then we calculate a matrix that we will call $\mathbf{\Sigma}^*$

$$\mathbf{\Sigma}^* := \frac{1}{T}\mathbf{S}\mathbf{S}^\top$$

Example 3.3. Now, let's apply this matrix to the datasets in Example 3.1 to see its effects on the allocation of the GMVP weights. First, transform the data matrix $\mathbf{X}_1$ and $\mathbf{X}_2$ via the spatial sign function. We then compute the $\mathbf{\Sigma}_i^* = \frac{1}{10}\mathbf{S}_i\mathbf{S}_i^\top$. The resulting matrices are bounded and well-behaved:

$$\mathbf{\Sigma}_1^* = \begin{bmatrix} 0.350 & -0.097 & -0.100 \\ -0.097 & 0.300 & -0.163 \\ -0.100 & -0.163 & 0.350 \end{bmatrix} \quad \mathbf{\Sigma}_2^* = \begin{bmatrix} 0.303 & -0.018 & -0.074 \\ -0.018 & 0.327 & -0.132 \\ -0.074 & -0.132 & 0.371 \end{bmatrix}$$

Notice that the trace of both matrices is exactly 1, an inherent geometric property of the SSCM that guarantees a finite-variance regime. More importantly, the massive 74.7 and 91.5 magnitude spikes seen in the standard $\mathbf{\Sigma_2}$ have been mathematically neutralized.

If we feed these transformed spatial matrices back into the GMVP optimizer, we obtain the following spatial weights:

$$\mathbf{w}_1^* = \begin{bmatrix} 0.285 \\ 0.375 \\ 0.340 \end{bmatrix} \qquad \mathbf{w}_2^* = \begin{bmatrix} 0.307 \\ 0.344 \\ 0.348 \end{bmatrix}$$

The results are immediately striking. The weights for the normal market ($\mathbf{w}_1^*$) remain stable and effectively identical to standard SCM. However, under the heavy-tailed market crash of Day 7, our transformed matrix completely prevents the optimizer from breaking down. Instead of demanding a highly leveraged, unstable 200% long/short allocation, $\mathbf{w}_2^*$ recognizes the Day 7 outlier as noise, stripping away its magnitude and returning a stable, balanced, long-only allocation that honors the true underlying structural relationships of the assets.

However, we are not done. Through the spatial sign function, we successfully counteracted the bias from the outlier. But our variances on the diagonal entries of $\mathbf{\Sigma}_i^*$ were affected. Hence, we

need to find a method to compensate for this. A suitable name for this matrix $\mathbf{\Sigma}^*$ would be a *scatter matrix.*

We clean the scatter matrix $\mathbf{\Sigma}^*$ by a 2-step process:

1. Transform it into a correlation matrix $\mathbf{C}$
2. Find suitable estimators for the variances, and form the proper SSCM via decomposition,

$$\mathbf{\Sigma}_{SSCM} := DCD$$

where $D := \text{diag}(\sigma_1^2, \ldots, \sigma_N^2)$

We formalise this below.

Step 1. (Finding $\boldsymbol{C}$). This step is simple. We compute $\mathbf{C}$ via the following decomposition:

$$\mathbf{C} = W\mathbf{\Sigma}^* W$$

where $\mathbf{W} := \text{diag}\left(\frac{1}{\Sigma_{11}^*}, \ldots, \frac{1}{\Sigma_{NN}^*}\right)$.

Step 2. (Estimating $\Sigma_{ii}^* = \sigma_i^2$). We want to be able to estimate the actual variances of each stock from our original data matrix $\mathbf{X}$ (pre spatial sign transformation). In addition, our estimator must be unbiased towards outliers. Hence, we suggest to use the *Median Absolute Deviation.*

Definition 3.4 (Median Absolute Deviation)**.** Let $x_i = (x_{i,1}, x_{i,2}, \ldots, x_{i,T})$ represent the vector of returns for the i-th asset across T observations, corresponding to the i-th row of the original data matrix $\mathbf{X}$. The Median Absolute Deviation (MAD_i) is formally defined as the median of the absolute deviations of the data points from their median, that is,

$$\text{MAD}_i = \text{median}\left(|x_{i,t} - \text{median}(x_i)|\right) \quad \text{for } t = 1, \ldots, T$$

In order to use the MAD as an asymptotically consistent estimator for the standard deviation σ_i under the assumption that the structural core of the data is normally distributed, it is multiplied by a scale factor of $k \approx 1.4826$ (derived from $1/\Phi^{-1}(0.75)$, where Φ is the cumulative distribution function of the standard normal distribution).

Consequently, the robust, outlier-resistant estimate for the asset's actual variance, Σ_{ii}, is given by:

$$\hat{\sigma}_i^2 = (1.4826 \cdot \text{MAD}_i)^2$$

And with this, we wrap up *Step 2* by obtaining (via MAD) estimators $\hat{\sigma}_i$ for $i = 1, \ldots, N$ and taking the matrix $\mathbf{C}$ computed in *Step 1*, and combining it to get our desired Spatial Sign Covariance Matrix

$$\mathbf{\Sigma}_{SSCM} := DCD$$

where $D := \text{diag}(\hat{\sigma}_1, \ldots, \hat{\sigma}_N)$

3.3 Cleaning the Spatial Sign Covariance Matrix

Now that we have a way to construct a more robust matrix to be used in portfolio optimization, we also want to do the same as we did for our covariance matrices in Chapter 2.2. We suggest an eigenvalue clipping scheme for the Spatial Sign Covariance Matrix.

1. *Compute* $\boldsymbol{\Sigma}^*$.

2. *Compute* $\boldsymbol{C}$. We compute the *correlation* matrix by normalising the diagonals to be 1. This allows us to fit the assumptions of the MP law and find the exact upper MP bound, given by $\lambda_+ = (1 + \sqrt{q})^2$. We then perform a spectral decomposition $\mathbf{C} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top$, where $\boldsymbol{\Lambda}$ is the diagonal matrix of sample eigenvalues and $\mathbf{V}$ is the orthogonal matrix of corresponding eigenvectors.

3. *Perform eigenvalue clipping on* $\boldsymbol{C}$. For all d eigenvalues where $\lambda_i > \lambda_+$, we isolate them as our signal. The $N - d$ eigenvalues are replaced by the average of all eigenvalues in the MP bulk. This is analogous to the method in Chapter 2.2.

4. *Apply Li's Bias Correction to the Signal.* While standard clipping leaves the d signal eigenvalues unchanged, Li et al. (2022) demonstrate that in high-dimensional regimes, these sample eigenvalues are biased upward by an order of $\mathcal{O}(1/N)$. To correct the true structural signals, we apply their moment-based correction. We compute the sample moments of the spectrum, subtract the theoretical limiting mean of the noise fluctuation (derived via their Central Limit Theorem), and solve the resulting moment equations for the true population eigenvalues, $a_1, \ldots, a_d$.

5. *Reconstruct the Final Covariance Matrix.* We form a new diagonal matrix $\boldsymbol{\Lambda}_{clean}$ containing our solved signal eigenvalues $a_1, \ldots, a_d$ and the averaged noise bulk $\bar{\lambda}_{noise}$. We rebuild the cleaned correlation matrix as $\mathbf{C}_{clean} = \mathbf{V}\boldsymbol{\Lambda}_{clean}\mathbf{V}^\top$. Finally, we reintroduce the real-world asset volatility by multiplying by our MAD-derived robust scale matrix $\mathbf{D}$, yielding our final estimator:
$$\boldsymbol{\Sigma}_{clean} := \mathbf{D}\mathbf{C}_{clean}\mathbf{D}$$

We claim that this SSCM construction performs better when the test data contains market noise – such as market crashes. Hence, we construct 3 GMVP portfolios:

1. *Control Period (January 2017 ∼ December 2017):* A period characterized by historically low market volatility.

2. *Global Financial Crisis (January 2008 – December 2009):* Represents an endogenous, structural market failure. This period is crucial for testing GMVP performance because asset correlations tend to converge toward 1.0 during severe systemic crashes, making standard diversification fail and severely testing the robustness of your covariance matrix.

3. *COVID-19 Crash (January 2020 – December 2020):* Captures a massive, exogenous market shock. Unlike the prolonged GFC, this period features an unprecedented, short-term spike in market noise followed by a rapid V-shaped recovery, testing how quickly your SSCM construction adapts to sudden regime changes without overfitting to the panic.

We use 3 years worth of data to test each test period.

Simulation. We conduct a simulation to show this. As for the training period, it is summarized in Table 6. For all the periods, the training period is kept to 3 years in order to make consistent

Scenario	**Training Period**	**Testing Period**	*Dimension* q
Control	`2015-01-01` $\sim$ `2017-12-31`	`2018-01-01` $\sim$ `2018-12-31`	$q = 0.607$
2008 Financial Crisis	`2007-01-01` $\sim$ `2009-12-31`	`2010-01-01` $\sim$ `2010-12-31`	$q = 0.529$
COVID-19	`2018-01-01` $\sim$ `2020-12-31`	`2021-01-01` $\sim$ `2021-12-31`	$q = 0.627$

Table 6: Training and Testing Periods Across Different Market Scenarios

the dimension q of our data matrix. Note that the training period here is selected such that the market crashes are *inside* the training data. The strength of SSCM that we are trying to test here is to be unaffected by the noise generated by market crashes in our training data. For each of the period, we test 5 portfolios generated by 4 covariance matrices

1. Uncleaned sample covariance matrix (SCM)
2. Clipped sample covariance matrix (Clipped SCM)
3. Uncleaned spatial sign covariance matrix (SSCM)
4. Cleaned spatial sign covariance matrix (SSCM) – clipped & shrinked via results in Li et al. (2022)

For each time period, we include the following:

1. A table with the portfolio's performance metrics (expected returns, Sharpe, etc.)
2. A histogram modeling the weight distribution predicted by Markowitz' framework, and a table with its metrics (Mean, Variance)

Before looking at the results we state our hypothesis. To show the effectivenss of SSCM, we want to show that we get a higher *sharpe* for our test portfolio against the test period. This is because we want to show that our model is able to "filter out" noise generated by market crashes. Secondly, we want to show that our weights are well-controlled. This can be seen by a low variance in our weight distribution. Of course, a dispersed weight allocation may not necessarily be a bad thing. But, an extremely dispersed portfolio goes against the philosophy of diversification, which is, for lack of a better term, the DNA of the Markowitz portfolio strategy.

Results. We notice from Table 7 that as desired, our cleaned SSCM portfolio resulted in the highest Sharpe ratio for both 2008 and 2020.

Next, we look at the weight distribution for all 4 portfolios. We first observe from Figure 6 that in general, the SCM has the most dispersed (indicated by a high variance) weight distribution. This is an indicator of the low-performance of the SCM. In general, the high performing portfolios, namely the clipped SCM and the cleaned SSCM has low variance (i.e. a diverse portfolio), which is desirable. If we look more closely in Table 8, we notice that for all three scenarios, the cleaned SSCM had the most diverse portfolio out of the four different ones.

This has showed us that the cleaned SSCM does provide a more robust portfolio that works well for periods coming out of financial crashes. However, there is one thing that is worth investigating, that is, the performance of SSCM for different kinds of financial crashes. There are crashes such as COVID where the market recovered farily quickly (in a few months). These are known as *flash crashes*. On the other hand, an incident like the 2008 Global Financial Crisis took around 2 or more years to recover. This is a larger time period of recovery. We have shown

Period	Portfolio	Ann. Return	Ann. Volatility	Sharpe Ratio	Max Drawdown
Control Period (2018)	SCM	-7.11%	14.56%	-0.489	-16.45%
	Clipped SCM	6.27%	11.16%	**0.562**	-13.18%
	Standard SSCM	-0.99%	13.96%	-0.071	-19.71%
	Cleaned SSCM (Li)	2.89%	13.34%	0.216	-16.65%
Global Financial Crisis (2008)	SCM	13.32%	9.51%	1.400	-8.20%
	Clipped SCM	14.26%	9.76%	1.462	-6.77%
	Standard SSCM	18.73%	10.64%	1.760	-5.74%
	Cleaned SSCM (Li)	19.10%	10.05%	**1.900**	-5.64%
COVID-19 Crash (2020)	SCM	11.23%	15.34%	0.732	-11.96%
	Clipped SCM	18.32%	10.82%	1.694	-5.13%
	Standard SSCM	10.15%	12.86%	0.789	-10.89%
	Cleaned SSCM (Li)	20.60%	9.56%	**2.156**	-5.97%

Table 7: Out-of-Sample Performance Across Market Scenarios

that SSCM does work well (to a certain extent) for general market crashes, but perhaps there is a much more robust method to deal with more specific cases. One downside of the SSCM is that it doesn't work the best as a general-use portfolio. This can be seen in our control portfolio, where the clipped SCM had a higher Sharpe compared to the cleaned SSCM. This brings me to a possible solution to create a more dynamic portfolio that works for general use.

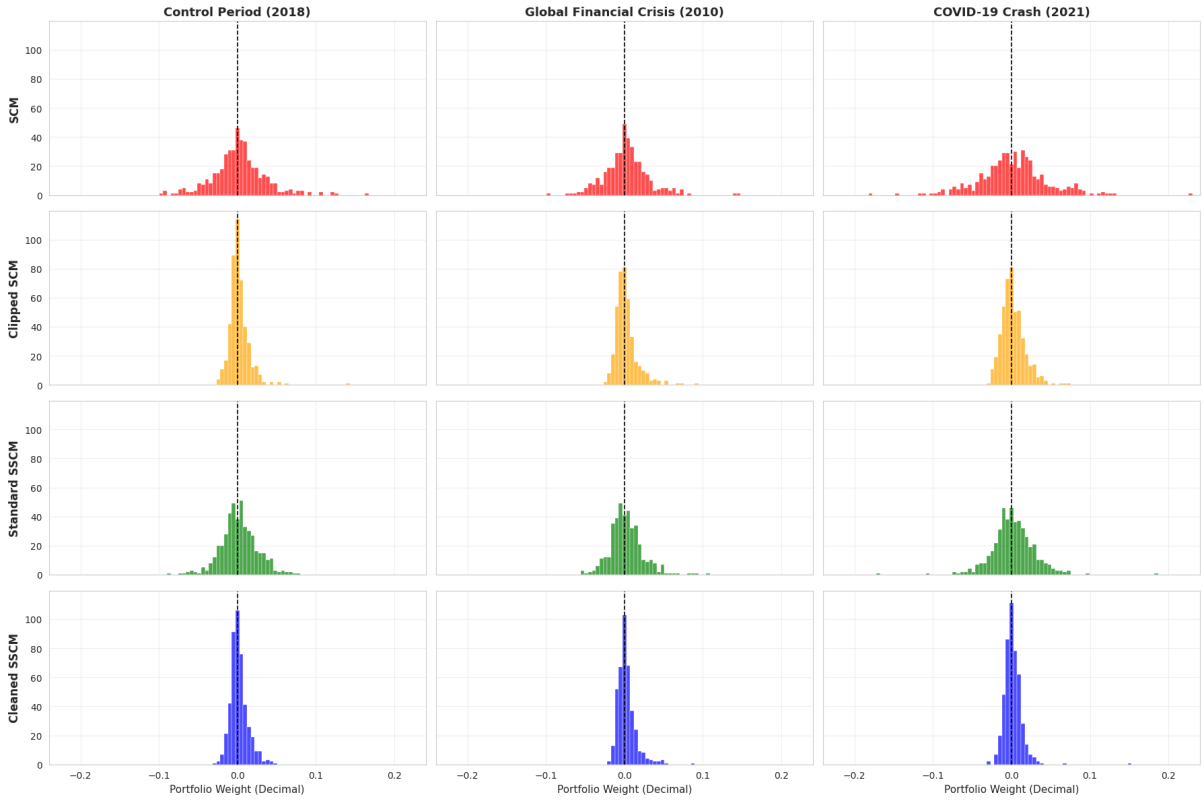


Figure 6: Weight distribution for the 4 different GMVP portfolios

3.4 A Convex Blend with the Gaussian-Cleaned Estimator

As seen in Table 7, the SSCM portfolio did not have the highest Sharpe for the control case. We suggest a simple yet powerful solution. In the spirit of covariance blending (See the linear shrinkage idea in Chapter 2.2, we suggest a convex blend between the cleaned SSCM, and a clipped SCM. We choose the clipped SCM as it performed the best in our cleaning test in Chapter 2.2.

Table 8: Weight distribution for the 4 different GMVP portfolios

Period	**Portfolio**	**Weight Mean**	**Weight Var.**
Control (2018)	SCM	0.002183	0.001126
	Clipped SCM	0.002183	0.000174
	Standard SSCM	0.002183	0.000547
	Cleaned SSCM	0.002183	**0.000123**
GFC (2010)	SCM	0.002506	0.000761
	Clipped SCM	0.002506	0.000201
	Standard SSCM	0.002506	0.000450
	Cleaned SSCM	0.002506	**0.000140**
COVID-19 (2021)	SCM	0.002114	0.001862
	Clipped SCM	0.002114	0.000218
	Standard SSCM	0.002114	0.000785
	Cleaned SSCM	0.002114	**0.000149**

The method. The method is essentially considering the following matrix, that we will call the *Dynamic Spatial Sign Covariance Matrix* (DSSCM)

$$\Sigma_{DSSCM} := \alpha\Sigma_{SSCM} + (1-\alpha)\Sigma_{Clipped}$$

As the name suggests, the choice of α is dynamically chosen. Given a training set of T observations (e.g. a 3-year window), we implement the following adaptive selection process. We first partition the training data matrix $\mathbf{X}$ into two segments. The first 80% of the observations represent the historical regime, while the final 20% act as a "validation window" to gauge the immediate market environment. Due to insufficient time for the project, we do not formalise this idea. However, this idea remains an interest to be done in the near future.

Concluding Remarks. Overall, our results reinforce the central message of the covariance-cleaning literature: naively estimated sample covariance matrices are dominated by noise, and correcting for this. Whether through eigenvalue shrinkage, phase-transition-aware debiasing, or robust statistics, yields portfolios that are both higher-performing and more stable. Future work could formalize the dynamic blending scheme we proposed, or test robustness across a broader range of market regimes and asset universes.

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